

Section 1: Preprocessing Decisions

- Used text fields: program, comments, llm_generated_program, llm_generated_university, term
- Used numeric fields: gpa, gre, gre_v, gre_aw
- Used categorical fields: degree, us_or_international
- Filtered to Accepted/Rejected only
- Removed duplicates by URL
- Created unified text format with field labels

Section 2: Model Input Template

Program: <value>

Comments: <value>

LLM Program: <value>

LLM University: <value>

Term: <value>

GPA: <value>

GRE: <value>

GRE V: <value>

GRE AW: <value>

Degree: <value>

Citizenship: <value>

Section 3: Training Configuration

- Model: DistilBERT-base-uncased
- Max length: 256
- Batch size: 32
- Epochs: 2
- Learning rate: 2e-5
- Optimizer: AdamW
- Device: CPU
- Training samples: 1,652 (10% of data)

Section 4: Evaluation Metrics

Test Accuracy: 61.96%

Test Precision: 61.69%

Test Recall: 68.49%

Test F1: 64.91%

Confusion Matrix:

[[1106, 903],

[669, 1454]]

Section 5: Comparison to Module 12

- Transformer can use text fields that the two-layer network couldn't
- Including comments and program text likely helps
- Pretrained model is more flexible but training is more complex
- Added complexity is justified by the richer input (text + structured)

Section 6: Ethics Reflection

- Bias in self-reported GradCafe data
- Missing information (research experience, letters, SOP)
- Should never be used for real decisions
- The disclaimer is essential
- Educational value: demonstrates ML pipeline end-to-end